Week 8, Lecture 2 WOD: Answer Key

Curvature by Hand (optional take-home)

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Questions with one correct answer are answered directly. Open-ended items (the "have you seen this before" part of 2c, the second half of 2d, and 3b) get discussion points instead. All numbers are computed in R when this file is knitted.

This WOD is optional and not collected (handed out Wed., Oct. 14, after it went unused in Week 7). Use the key for questions or post it. If a few minutes open up, 2d is the part worth going over in class.

Part 1: Pika haypiles (Poisson)

1a.

> Find $\text{NLL}'(\lambda)$ and $\text{NLL}''(\lambda)$, where $\text{NLL}(\lambda) = n\lambda - (\sum y_i)\log\lambda$.

Answer. $\text{NLL}'(\lambda) = n - \dfrac{\sum y_i}{\lambda}$ and $\text{NLL}''(\lambda) = \dfrac{\sum y_i}{\lambda^2}$.

Why. $\frac{d}{d\lambda}\log\lambda = 1/\lambda$, and $\frac{d}{d\lambda}\lambda^{-1} = -\lambda^{-2}$; the two minus signs cancel. $\sum y_i$ is a constant (the data are fixed).

Watch for. Treating $\sum y_i$ as something to differentiate. Sign errors: NLL'' must be positive (the NLL has a minimum). Students who differentiate the log-likelihood instead get $-\sum y_i/\lambda^2$, which is fine as long as they flip the sign before taking the square root.

1b.

> Evaluate $\text{NLL}''(\hat\lambda)$ at $\hat\lambda = 3$ for the six plots, and again for 24 plots with the same mean. Give the SE for each.

Answer. 6 plots: $18/9 = 2$, $\text{SE} = 1/\sqrt{2} \approx 0.71$. 24 plots: $72/9 = 8$, $\text{SE} = 1/\sqrt{8} \approx 0.35$.

Why. Plug $\sum y_i$ (18 or 72) and $\lambda = 3$ into 1a. The first matches the `optim()` Hessian shown in the Week 7 Monday lecture.

Watch for. Using $\sum y_i = 18$ for the 24-plot case. Reporting the curvature as the SE (forgetting $1/\sqrt{\cdot}$).

1c.

> Substitute $\hat\lambda = \bar y$ into your NLL'' and simplify to a general formula for $\text{SE}(\hat\lambda)$. What happens to the SE when n quadruples?

Answer. $\text{NLL}''(\hat\lambda) = \dfrac{n\hat\lambda}{\hat\lambda^2} = \dfrac{n}{\hat\lambda}$, so $\text{SE}(\hat\lambda) = \sqrt{\hat\lambda/n}$. Quadrupling n halves the SE.

Why. $\sum y_i = n\bar y = n\hat\lambda$. The SE scales as $1/\sqrt{n}$. (Since a Poisson variance equals its mean, this is the familiar $\sqrt{\text{variance}/n}$ standard error of a mean.)

Watch for. Saying the SE drops by a factor of 4. That's the curvature (4 times steeper); the SE drops by $\sqrt{4} = 2$.

Part 2: Frog detection (binomial)

2a.

Find $\text{NLL}''(p)$ for $\text{NLL}(p) = -K \log p - (N - K) \log(1 - p)$.

Answer. $\text{NLL}'(p) = -\dfrac{K}{p} + \dfrac{N-K}{1-p}$ and $\text{NLL}''(p) = \dfrac{K}{p^2} + \dfrac{N-K}{(1-p)^2}$.

Why. $\frac{d}{dp}\log(1-p) = -\frac{1}{1-p}$ (chain rule); then $\frac{d}{dp}(1-p)^{-1} = (1-p)^{-2}$ (chain rule again, the two minus signs cancel). Both terms of NLL'' are positive.

Watch for. Dropping the chain-rule minus sign, which leaves $\frac{K}{p^2} - \frac{N-K}{(1-p)^2}$ (can go negative, a giveaway). The first derivative should match Lab 5 Act 1, 2a (with the sign flipped if they used the log-likelihood there).

2b.

Evaluate $\text{NLL}''(\hat{p})$ and the SE for $K = 12$, $N = 30$ and for $K = 120$, $N = 300$. Check: one of your curvatures should be 125.

Answer. 3 wetlands: $\text{NLL}'' = 125$, SE ≈ 0.089. 30 wetlands: $\text{NLL}'' = 1250$, SE ≈ 0.0283.

Why. For 3 wetlands: $12/0.16 + 18/0.36 = 75 + 50 = 125$. Ten times the data gives ten times the curvature and an SE smaller by $\sqrt{10} \approx 3.2$, matching the narrower dashed curve in Lab 5 Act 1, Part 3.

Watch for. Arithmetic with 0.4^2 and 0.6^2. Students who got 125 for the first may still forget to scale K and N together for the second.

2c.

Substitute $\hat{p} = K/N$ and simplify to a general formula for $\text{SE}(\hat{p})$. Have you seen this formula before?

Answer. $\text{NLL}''(\hat{p}) = \dfrac{N}{\hat{p}(1-\hat{p})}$, so $\text{SE}(\hat{p}) = \sqrt{\dfrac{\hat{p}(1-\hat{p})}{N}}$. This is the standard error of a proportion from intro stats.

Why. With $K = N\hat{p}$ and $N - K = N(1-\hat{p})$: $\frac{N\hat{p}}{\hat{p}^2} + \frac{N(1-\hat{p})}{(1-\hat{p})^2} = \frac{N}{\hat{p}} + \frac{N}{1-\hat{p}} = \frac{N}{\hat{p}(1-\hat{p})}$.

Discussion point. The "canned" formula from intro stats is the quadratic approximation to this likelihood. That's the payoff of the whole WOD: familiar formulas fall out of the curvature of a likelihood, and the same recipe works for models with no textbook formula.

Watch for. Getting stuck on the common denominator in the last step; the algebra is short once K and $N - K$ are rewritten in terms of $\hat{p}$.

2d.

The Week 7 Monday WOD (*Reading the Interval*) had 2 chytrid-positive frogs out of 20. Build $\hat{p} \pm 1.96\,\text{SE}$. What's wrong with it? Compare it with the range you read off the NLL curve in that WOD.

Answer. SE $= \sqrt{0.1 \times 0.9/20} \approx 0.067$, so the interval is -0.031 to 0.231. The lower end is a **negative prevalence**, which is impossible. The Week 7 WOD's 2-unit range was about 0.02 to 0.28: entirely positive, and lopsided (longer on the high side).

Why. The quadratic approximation is symmetric by construction, but this NLL curve isn't: $\hat{p}$ sits close to the boundary at 0, where the curve rises very steeply. With few positives the parabola is a poor stand-in for the real curve. The likelihood-based range follows the curve and so respects the boundary.

Watch for. Students who report "0 to 0.23" without comment (truncating hides the problem rather than explaining it). Good answers connect this to the Week 7 Monday lecture: the Wald and likelihood-ratio intervals agree when the curve is close to a parabola (lots of data, $\hat{p}$ away from 0 or 1) and disagree here.

Part 3: Check it in R

3a.

> *(Recommended: no AI.)* Write an NLL function for $K = 12$, $N = 30$ using `dbinom()`, and minimize it with `optim(..., method = "Brent", lower = 0.001, upper = 0.999, hessian = TRUE)`. Does the Hessian match 2b?

Answer. About 125 (for example 125.001). Yes, it matches.

```
nll_frog <- function(p) -dbinom(12, size = 30, prob = p, log = TRUE)
fit <- optim(0.5, nll_frog, method = "Brent", lower = 0.001, upper = 0.999,
             hessian = TRUE)
fit$par
```

```
## [1] 0.4
```

```
fit$hessian
```

```
##          [,1]
## [1,] 125.0012
```

Why. `optim()` approximates the second derivative by finite differences, so it agrees with the calculus up to small numerical error. The binomial coefficient term doesn't depend on p, so including it (via `dbinom()`) doesn't change the curvature.

Watch for. Forgetting the minus sign (maximizing instead of minimizing gives the MLE at a boundary and a nonsense Hessian). Omitting `method = "Brent"`, which triggers a warning for one-parameter problems.

3b.

> Where did you use AI, if anywhere, and what did you have to check or fix? If you didn't use it, what was hardest?

Answer (discussion points). No single correct answer. Useful responses name a specific step (the chain rule in 2a, the algebra in 2c, setting up `optim()`). If several students say the algebra in 2c was hardest, a quick board walkthrough is worth the time.